

# PREDICTIVE FAILURE CONTROL Of Stepper Motors Using PID

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Stepper motors are a mainstay in precision control systems, valued for their simple open-loop operation and high torque at low speeds. Commonly found in CNC machines, 3D printers, and robotics, they offer reliable motion control—yet their open-loop nature leaves them vulnerable to missed steps, overheating, and mechanical wear. Introducing a closed-loop PID (proportional-integral-derivative) control system significantly improves performance, enabling real-time monitoring to predict and prevent failures.

This system integrates a PID algorithm with predictive and

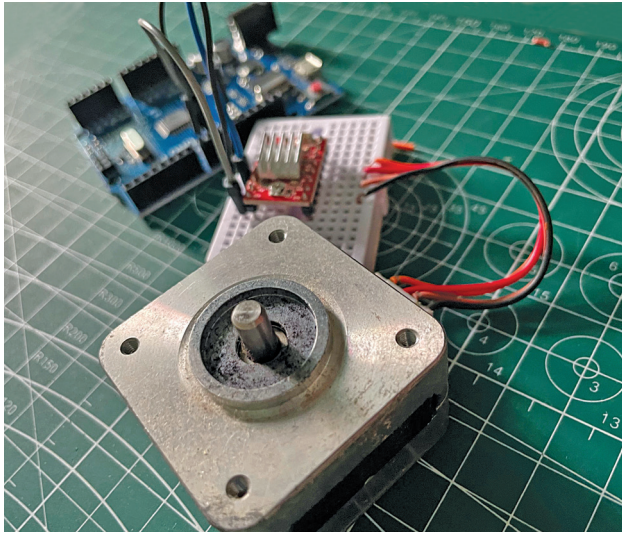


Fig. 1: Author's prototype

preventive fault detection, implemented on an Arduino microcontroller and simulated using Proteus. Before exploring the implementation, it helps to revisit the core principles of PID control and the formula used. The complete list of required components is provided in the

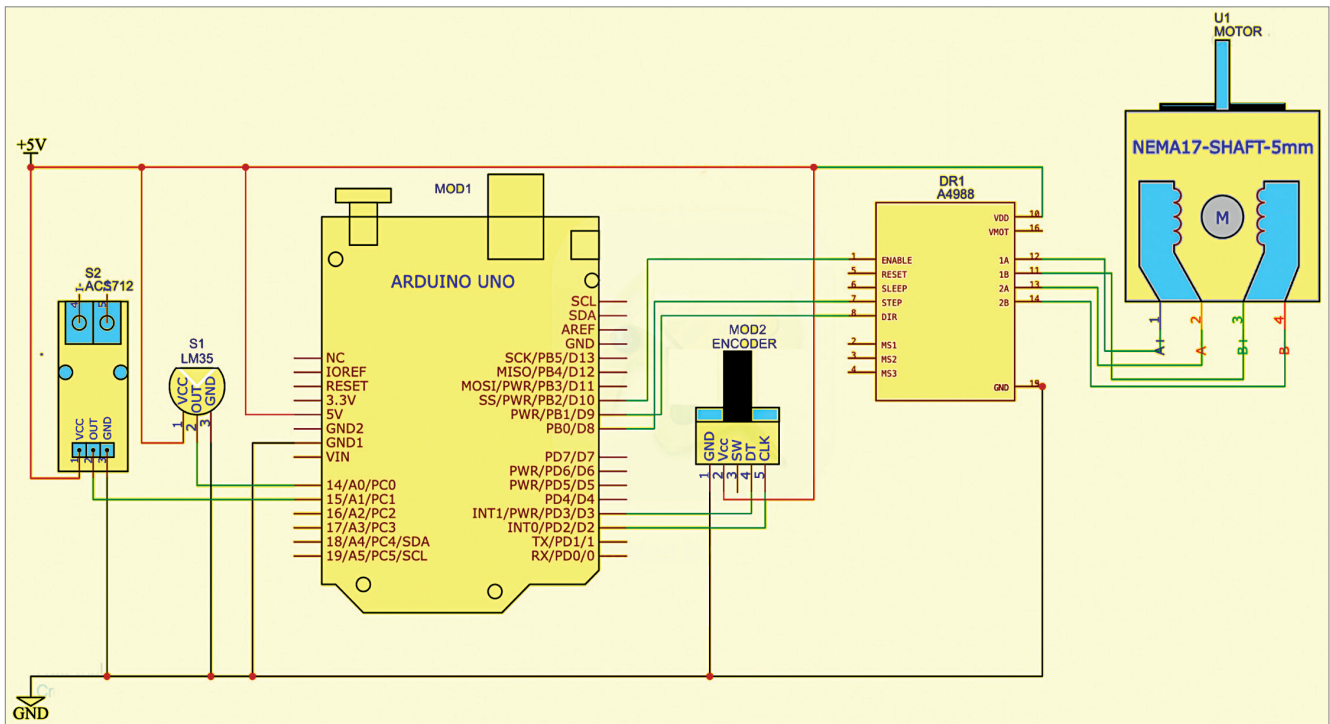


Fig. 2: Circuit diagram